

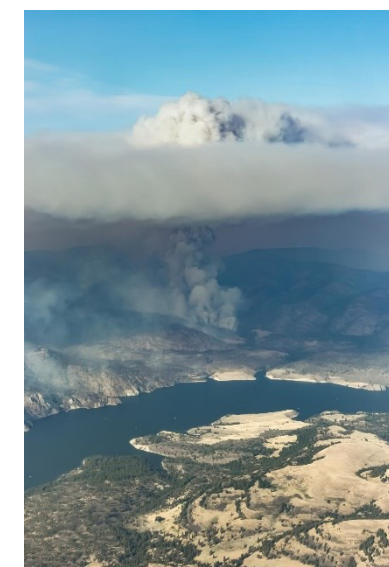
3D Animation of Wildfire Plumes



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Abstract



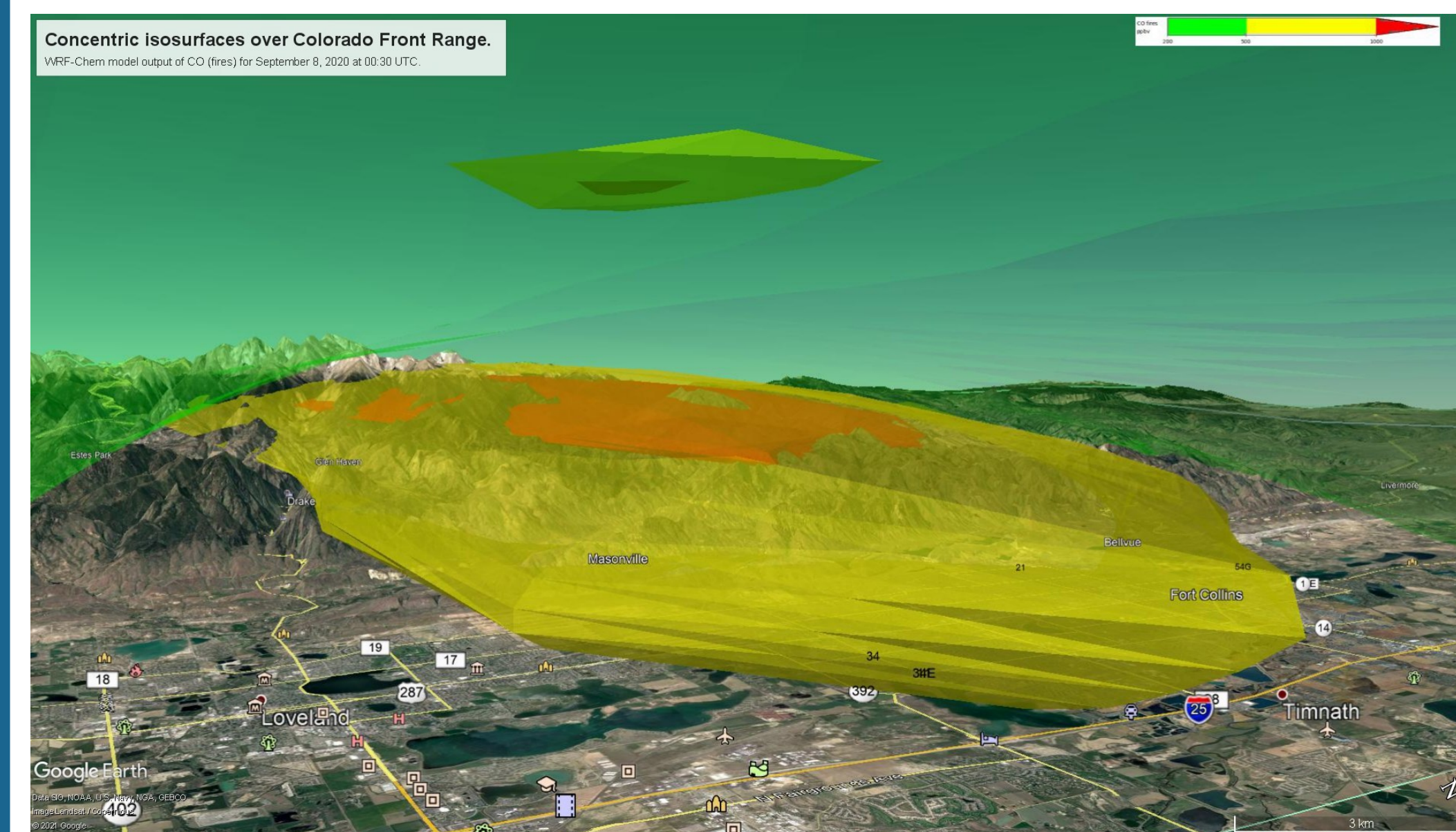
ACOM flew the NASA DC-8 over the Williams Flats fire on August 3, 6, 7, and 8, 2019. Smoke plumes from this wildfire reached unusually high, modeled by WRF-Chem at 10 km (32,800 feet). Photos by Sam Hall and Rebecca Hornbrook.



Smoke Plumes from Wildfires

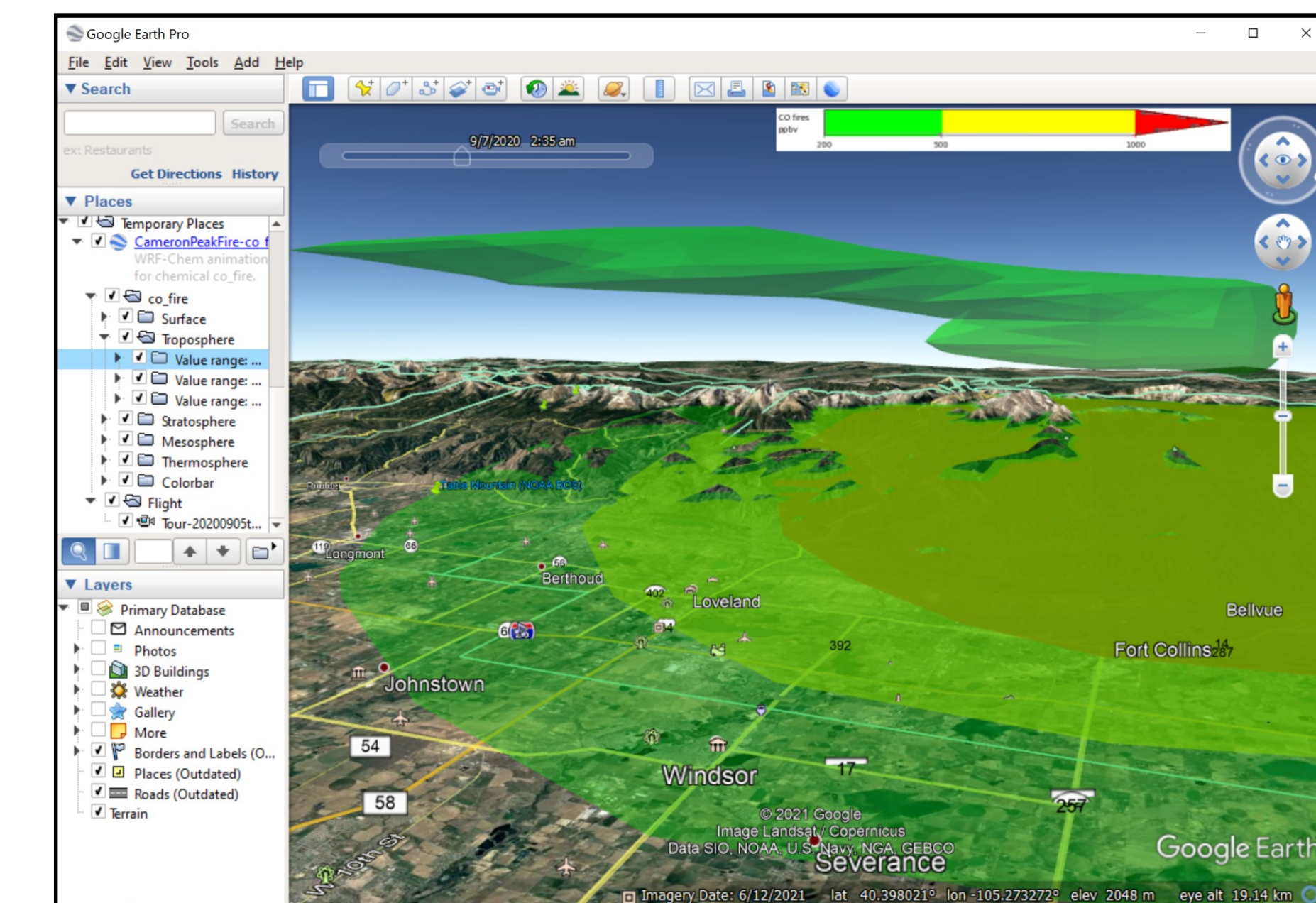
During the late summer and early fall of 2020, Colorado experienced several large wildfires. The Atmospheric Chemistry Observations & Modeling laboratory (ACOM) at the National Center for Atmospheric Research (NCAR) runs the atmospheric chemistry model WRF-Chem to track emissions from these fires and simulate the spread of aerosol particles and chemicals. We have developed a formulation of the Marching Cubes algorithm to create 3D isosurfaces at several chemical concentrations, and we use Google Earth as a rendering and visualization platform to display and analyze these plumes of smoke. Our Python scripts generate a multi-day animation of the wildfire, with a KMZ "tour" that flies around the active base of the fire while it is burning.

3D Isosurfaces



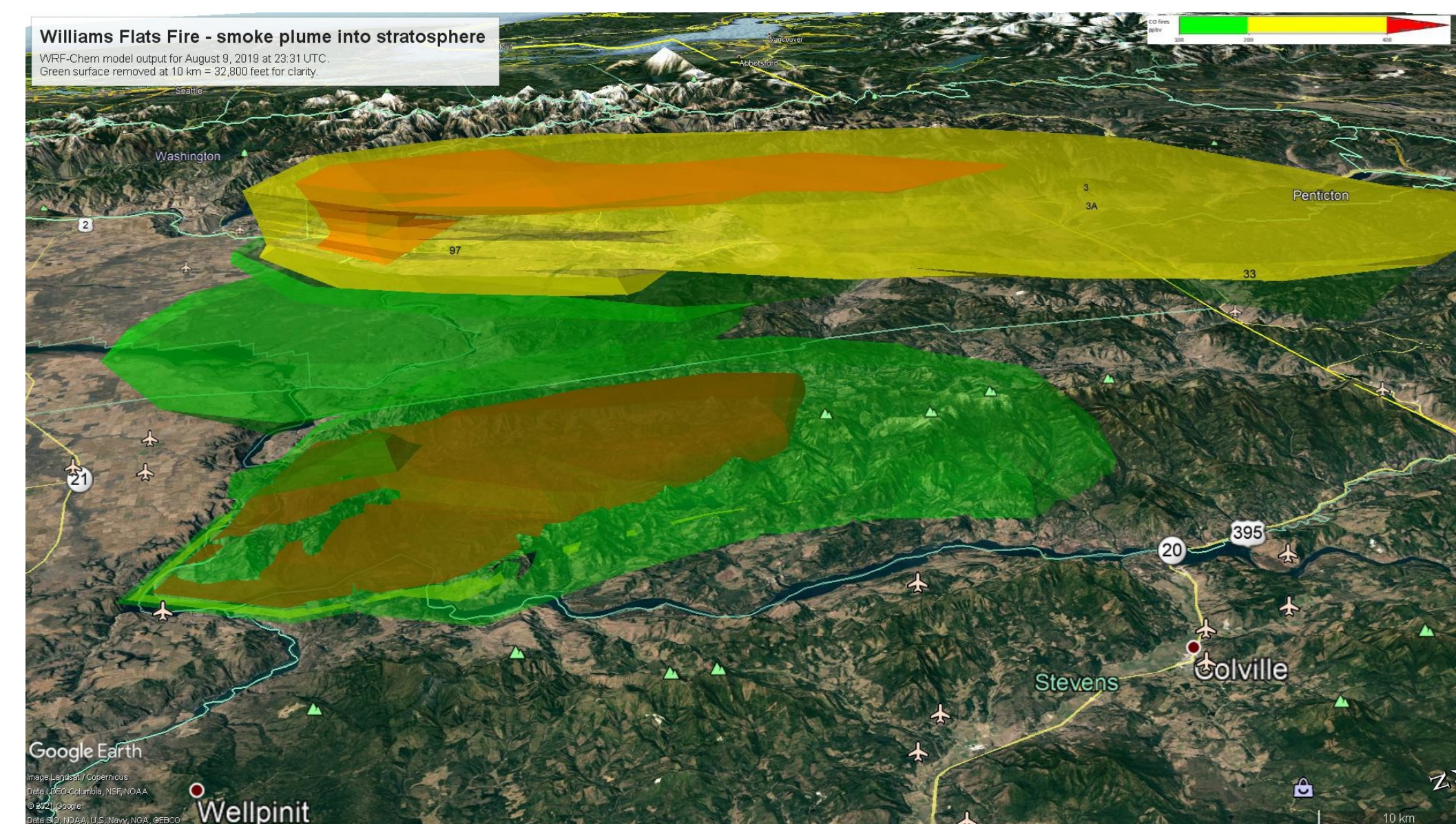
ACOM uses our formulation of the Marching Cubes algorithm to generate 3D isosurfaces for each hourly step of WRF-Chem model output. Here we see a plume of smoke from the Cameron Peak fire stretching over Loveland, Colorado. This view is from inside the green isosurface (200 ppbv), looking at the yellow surface (500 ppbv), and red surface (1000 ppbv) colored as orange within the yellow envelope. The time is September 8, 2020 at 00:30 UTC.

Google Earth



Google Earth makes an excellent viewer for the 3D rendering of smoke clouds above a realistic earth surface. We use the controls in the left sidebar to classify and display the three isosurfaces by atmospheric layer and value. The desktop application also comes with a large set of earth annotations to clarify the view.

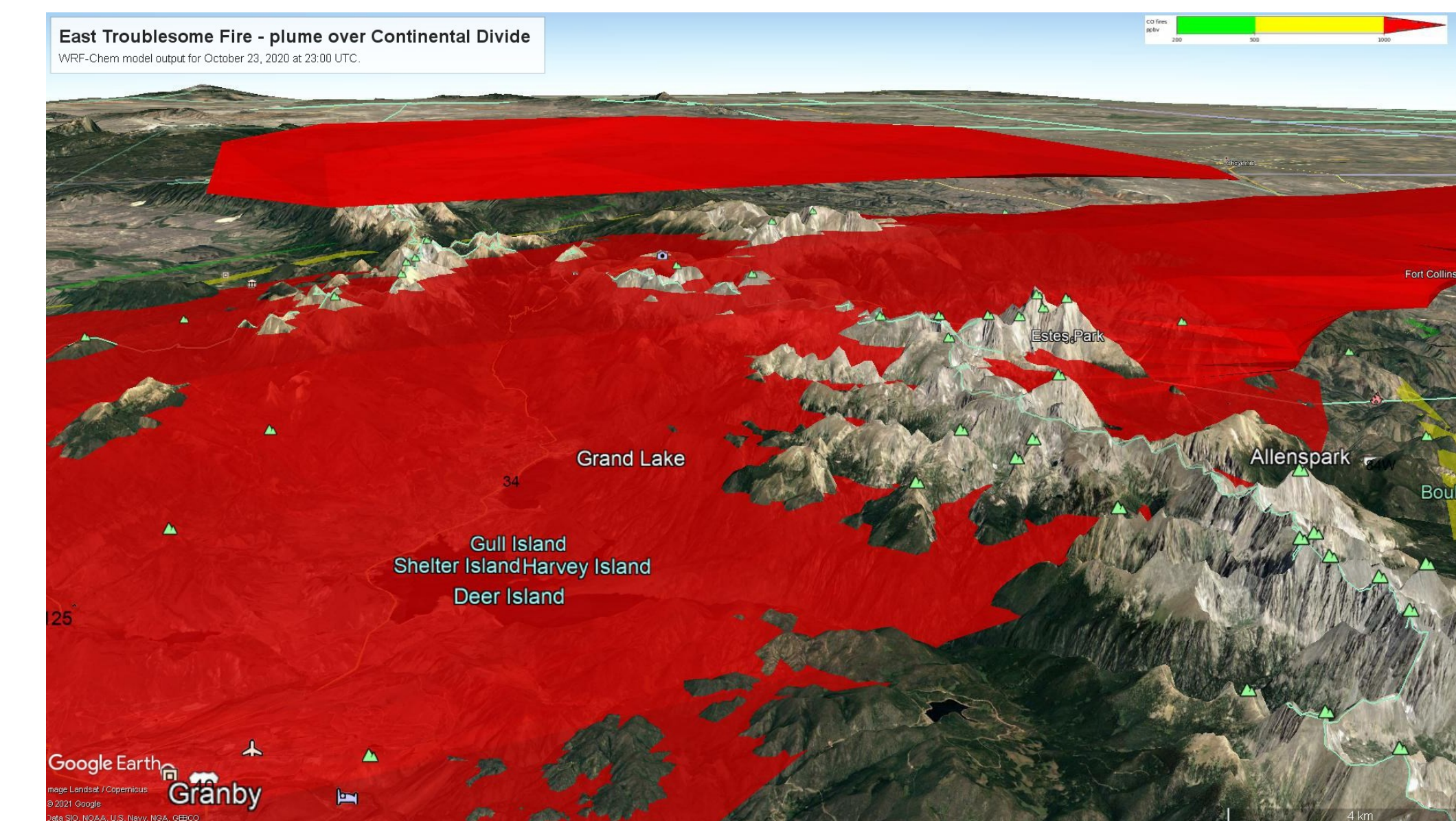
Williams Flats Fire 2019, Washington



Plumes into the stratosphere

The ACOM visualization sets a numerical boundary at 10 km above sea level, which roughly corresponds to the jet stream and strong upper-level winds. Here we see a dense smoke plume from Williams Flats rising into this upper layer and being blown to the north-northeast by high winds. The green isosurface has been removed above 10 km for clarity.

Interaction With Terrain



3X vertical exaggeration

We have set the vertical exaggeration in Google Earth and our COLLADA files to 3x in order to emphasize the vertical structure of the atmosphere. This figure shows a dense plume of smoke from the East Troublesome fire blowing over the Continental Divide into Estes Park and toward Ft. Collins on October 23, 2020 at 23:00 UTC. Although the 12 km grid resolution of WRF-Chem cannot represent Rocky Mountain National Park accurately, Google Earth loads higher-resolution terrain when zooming in; the resulting image provides greater fidelity to the alpine geography.

Download:

Acknowledgements:

We have posted several .kmz archives for download at the following web site: <https://www.acom.ucar.edu/webt/wrf-chem/colorado-wildfires/>. We encourage participants in Vis 2021 to download our KMZ archives, experiment with Google Earth, and give us their comments to improve ACOM's wildfire animation system. E-mail: drews@ucar.edu

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